Hochschild Homology of the b-Pseudodifferential Symbol Algebra

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Abstract

Let X be a compact n-manifold with embedded boundary and let $E \to X$ be a complex vector bundle. We give an expository account of the continuous Hochschild homology of the Benameur–Nistor Laurent complete-symbol algebra of the b-groupoid,

$$\mathcal{A}_{b,L}(X;E) = \mathcal{O}(X)\left(\Psi^{\infty}(\mathcal{G}_b;E)/\Psi^{-\infty}(\mathcal{G}_b;E)\right),$$

where $\mathcal{O}(X)$ consists of functions with finite Laurent singularities at the boundary and the algebra carries the topologically filtered completion of Benameur–Nistor. The result,

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}({}^bS^*X \times S^1_\sigma;\mathbb{C}),$$

is the b-groupoid specialization of the Benameur–Nistor theorem on Laurent complete symbols. The circle S^1_σ records symbol order and H_L^* is Laurent de Rham cohomology at the boundary. We give a self-contained computation of the E^1- and E^2-pages of the order spectral sequence and check the hypotheses for the b-groupoid; convergence and the continuous Hochschild–Kostant–Rosenberg theorem are quoted. A bidegree count shows that E^2 is concentrated on one row, so the spectral sequence degenerates and, given convergence, the displayed isomorphism is canonical; only its splitting into four interior and boundary summands involves choices. We also describe an alternative Čech–de Rham organization, which depends on a quoted localization theorem for Hochschild chains, and a heuristic Rees–Fedosov reorganization that is not an independent proof. Finally we separate this result from the Hochschild homology of an unreduced operator algebra, which depends on the chosen residual ideal and completion.

1 The precise statement

The phrase *the algebra generated by b-pseudodifferential operators* is ambiguous. At least four choices affect Hochschild homology: the allowed operator orders, the allowed powers of a boundary defining function, the residual ideal, and the locally convex completion. We use the standard algebra for which the homology has a finite-dimensional geometric description. It is the b-groupoid specialization of the Laurent symbol algebras studied in [2, 1].

Status of the result. Theorem 1.1 below is the b-groupoid case of the Benameur–Nistor theorem on Laurent complete symbols [2, 1]; it is not a new theorem. What these notes add is a self-contained computation of the E^1- and E^2-pages of the order spectral sequence in the b-setting (Sections 4–7), an elementary proof that the spectral sequence degenerates at E^2 (Section 8), and a check of the hypotheses for the b-groupoid. Two inputs are quoted from Benameur–Nistor: convergence of the order spectral sequence for the completed continuous Hochschild complex, and the continuous, globalized Hochschild–Kostant–Rosenberg theorem identifying E^1.

For one boundary hypersurface, the standard b-groupoid has underlying set

$$\mathcal{G}_b = X^\circ \times X^\circ \sqcup \bigsqcup_{H \in \pi_0(\partial X)} H \times H \times \mathbb{R}.$$

On a boundary component, composition adds the last coordinate. Its smooth structure is fixed by the normal ratio $t = \log(x/x')$, and its Lie algebroid is $A(\mathcal{G}_b) = {}^bTX$. For corners, one uses the standard groupoid with an $\mathbb{R}^j$-factor over each open codimension-j face [1].

Let $\mathcal{O}(X)$ be the ring generated by $C^\infty(X)$ and x_H^{-1} for all boundary hypersurfaces H. Benameur and Nistor define

$$\mathcal{A}_{b,L}(X;E) := \mathcal{O}(X)\left(\Psi^{\infty}(\mathcal{G}_b;E)/\Psi^{-\infty}(\mathcal{G}_b;E)\right)$$

with their topologically filtered topology and completed Hochschild complex [2, 1]. Here $\Psi^{\infty}(\mathcal{G}_b;E)$ denotes the $\mathcal{G}_b$-pseudodifferential operators acting on sections of E (their kernels are distributional

sections over $\mathcal{G}_b$ of the pull-backs of E and E^* by the range and source maps); we write E throughout. The vector representation identifies this with the usual properly supported Laurent b-complete symbols. In particular, one may write its underlying filtered union heuristically as

$$\frac{\bigcup_{m,\ell} x^{-\ell}\Psi_b^m(X;E)}{\bigcup_\ell x^{-\ell}\Psi_b^{-\infty}(X;E)},$$

but this quotient notation does not by itself specify the topology used in the theorem.

Changing the defining function from x to $x' = e^f x$ gives

$$(x')^{-\ell} = e^{-\ell f} x^{-\ell}.$$

Hence the change only multiplies a Laurent generator by a smooth invertible function, and the algebra, its filtration, and the answer below are canonically independent of this choice. For corners one allows an arbitrary finite Laurent monomial $\prod_H x_H^{-\ell_H}$, one factor for each boundary hypersurface.

Set $Y = {}^bS^*X = ({}^bT^*X \setminus 0)/\mathbb{R}_+$ and $Z = Y|_{\partial X}$. Thus $\partial Y = Z$. The symbol-order circle is denoted S^1_σ.

Theorem 1.1 (Hochschild homology; b-groupoid case of Benameur–Nistor)**.** *Assume the Benameur–Nistor convergence theorem for the order spectral sequence of $\mathcal{A}_{b,L}(X;E)$ (Section 8). Then for every $k \geq 0$ the induced filtration of $\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E))$ has exactly one nonzero graded piece, and there is a canonical isomorphism*

$$\boxed{\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}(Y \times S^1_\sigma;\mathbb{C}).} \tag{1}$$

The answer is independent of E. If X has one embedded boundary hypersurface, then

$$\begin{aligned}\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong\ & H^{2n-k}(Y) \oplus H^{2n-k-1}(Y) \\ & \oplus H^{2n-k-1}(Z) \oplus H^{2n-k-2}(Z),\end{aligned} \tag{2}$$

where a cohomology group in negative degree is zero. In particular the homology vanishes for $k > 2n$.

The isomorphism (1) is canonical once convergence is granted; the Künneth decomposition for S^1_σ is canonical as well. The four-summand formula (2) uses one choice: a splitting of the Laurent face sequence (a collar, defining function, and cutoff; Proposition 7.1). The subspace $H^{2n-k}(Y \times S^1_\sigma)$ and the boundary quotient $H^{2n-k-1}(Z \times S^1_\sigma)$ are canonical.

Remark 1.2 (Comparison with Melrose–Nistor). Melrose and Nistor [8] compute the Hochschild and cyclic homology of the algebra of *cusp* pseudodifferential operators on a compact manifold with boundary. This is the full operator algebra, including its residual ideal, not a symbol quotient. They interpret the index as a Hochschild 1-cocycle and obtain an extension of the Atiyah–Patodi–Singer formula, whose boundary term is an eta invariant of the suspended algebra of the boundary. Their work does not treat the b-calculus. The present theorem concerns only the Laurent symbol quotient of the b-calculus. The b-calculus analogue of the Melrose–Nistor computation is not addressed here: the Hochschild homology of the full operator algebra $\Psi_b^{\mathbb{Z}}(X;E)$ and the connecting map to the symbol algebra. Section 12 explains why that problem depends on the choice of residual ideal and completion.

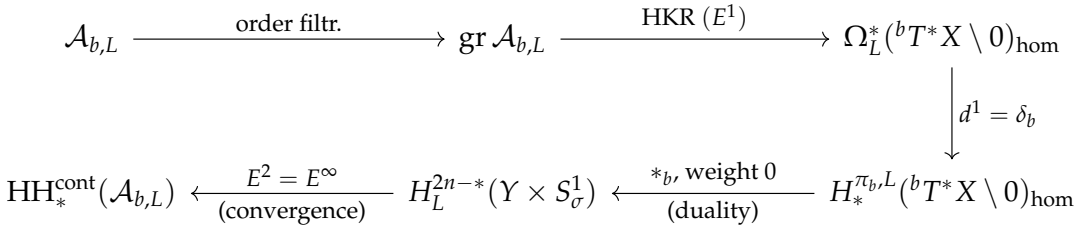


Figure 1: The computation passes from the filtered noncommutative algebra to its commutative associated graded (E^1), then from Poisson homology to Laurent de Rham cohomology (E^2). Only the weight-zero part survives, which forces $E^2 = E^\infty$; convergence is the quoted input.

2 Continuous chains and the two filtrations

The topology is part of the theorem. For a topologically filtered algebra A, Benameur–Nistor do *not* take the completed projective tensor power of the whole locally convex space: multiplication need not be jointly continuous for that topology. In their notation the degree-k chain space is

$$C_k^{\mathrm{cont}}(A) := \varinjlim_N \big(F^N_{N,N}A\big)^{\widehat{\otimes}_\pi(k+1)}. \tag{3}$$

(Benameur–Nistor denote this chain space by $H_k(A)$; we avoid that notation, which suggests homology.) Here $F^N_{N,N}A$ is the diagonal cofinal family of filtration pieces of [1]; equivalently, one first takes the projective completion in the decreasing symbol-order direction and only then enlarges the support and Laurent bounds by an inductive limit. We apply this with $A = \mathcal{A}_{b,L}$. Whenever a tensor product below carries no decoration it denotes an elementary tensor in the dense filtered subcomplex of (3), not an ordinary completed tensor power of the whole algebra.

For a chain $a_0 \otimes \cdots \otimes a_k$, the Hochschild boundary is

$$\begin{aligned} b(a_0 \otimes \cdots \otimes a_k) = &\sum_{j=0}^{k-1} (-1)^j a_0 \otimes \cdots \otimes a_j a_{j+1} \otimes \cdots \otimes a_k \\ &+ (-1)^k a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}. \end{aligned}$$

Multiplication is continuous on fixed filtration pieces, so b extends to the completed complex and satisfies $b^2 = 0$.

Laurent order and symbol order play different roles. The first is built into $\mathcal{O}(X)$; symbol order produces the spectral sequence. On every fixed piece in (3), filter elementary tensors by

$$\operatorname{ord}(a_0) + \cdots + \operatorname{ord}(a_k) \le p,$$

take the projective closure there, and then pass through the same projective/inductive limits. Denote the resulting subspace by $\mathcal{F}_pC_k$. Then $b(\mathcal{F}_pC_k) \subseteq \mathcal{F}_pC_{k-1}$. Exhaustivity, completeness, and the convergence hypotheses are those verified in the Benameur–Nistor topologically filtered theorem. Borel quantization explains asymptotic completeness of symbols, but it is not by itself a convergence proof for an arbitrarily completed Hochschild complex.

Discussion. **Why divide by smoothing operators?** The principal-symbol filtration sees an operator only through its full asymptotic symbol. Smoothing kernels form the kernel of that passage. Keeping them gives an extension problem, treated separately in Section 12; it is not a harmless change of notation.

3 The b-cotangent geometry

The Lie algebra of b-vector fields is

$$\mathcal{V}_b(X) = \{V \in C^\infty(X; TX) : V \text{ is tangent to } \partial X\}.$$

It is the space of smooth sections of a vector bundle ${}^bTX \to X$. Near a boundary point, choose product coordinates

$$(x, y_1, \dots, y_{n-1}).$$

Then

$$\mathcal{V}_b(X) = C^\infty(X)\text{-span}\{x\partial_x, \partial_{y_1}, \dots, \partial_{y_{n-1}}\}.$$

The dual local frame of ${}^bT^*X$ is

$$\frac{dx}{x}, \quad dy_1, \dots, dy_{n-1}.$$

Thus a b-covector has the form

$$\xi\frac{dx}{x} + \eta \cdot dy.$$

On ${}^bT^*X$ the tautological b-one-form and its differential are

$$\alpha_b = \xi\frac{dx}{x} + \eta \cdot dy, \qquad \omega_b = d\alpha_b = d\xi \wedge \frac{dx}{x} + d\eta \wedge dy.$$

The form ω_b is nondegenerate as a b-two-form. Its inverse is the b-Poisson tensor

$$\pi_b = x\partial_x \wedge \partial_\xi + \sum_j \partial_{y_j} \wedge \partial_{\eta_j}.$$

This is the geometric bracket appearing in the first nonzero differential of the Hochschild spectral sequence.

Proposition 3.1 (Anchor bridge). *(i) The anchor $a : {}^bTX \to TX$ becomes an isomorphism of Lie algebroids after adjoining Laurent coefficients:*

$$\mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma({}^bTX) \xrightarrow{a} \mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma(TX),$$

an isomorphism of $\mathcal{O}(X)$-modules preserving brackets.

(ii) There is a (noncanonical) isomorphism of smooth vector bundles ${}^bTX \cong TX$. Consequently $Y = {}^bS^*X \cong S^*X$ as sphere bundles over X, by a diffeomorphism carrying Z onto $S^*X|_{\partial X}$, and

$$H_L^*({}^bS^*X \times S^1_\sigma) \cong H_L^*(S^*X \times S^1_\sigma).$$

The isomorphism in (ii) is not induced by the anchor: the dual anchor $T^*X \to {}^bT^*X$ does not identify the homogeneous symbol spaces.

Proof. (i) Locally the anchor sends $x\partial_x$ to the ordinary vector field $x\partial_x$ and fixes ∂_{y_j}. Its localized inverse uses the Laurent coefficient x^{-1}: $\partial_x \mapsto x^{-1}(x\partial_x)$. The anchor of a Lie algebroid preserves brackets, hence so does its Laurent localization.

(ii) In a collar $[0, \varepsilon)_x \times \partial X$ choose a smooth function $h(x) > 0$ with $h = 1$ near $x = 0$ and $h(x) = x$ for $x \geq \varepsilon/2$. Define $\Phi : {}^bTX \to TX$ by $\Phi(x\partial_x) = h(x)\partial_x$, $\Phi(\partial_{y_j}) = \partial_{y_j}$ in the collar, and $\Phi = a$ outside it. Then Φ is smooth and fiberwise invertible, since $h > 0$ and a is invertible over X°.

Its inverse transpose induces a diffeomorphism of cosphere bundles ${}^bS^*X \to S^*X$ over the identity of X, hence carries the restriction to ∂X onto the restriction to ∂X. A boundary defining function of S^*X pulls back to one of Y, so the diffeomorphism identifies the Laurent de Rham complexes.

Why not more: in the coordinates above the dual anchor is $\xi\,dx + \eta\cdot dy \mapsto (x\xi)\,dx/x + \eta\cdot dy$, so it kills the conormal direction at ∂X and does not map $T^*X\setminus 0$ into ${}^bT^*X\setminus 0$. For instance, the homogeneous function $\xi_b/|\zeta_b|$ pulls back to $x\xi/\sqrt{x^2\xi^2+|\eta|^2}$, which is not Laurent-smooth near $x=\eta=0$. So no chain-level isomorphism of homogeneous Laurent symbol complexes is induced by the anchor. □

4 The associated graded algebra

Filter $\mathcal{A}_{b,L}$ by operator order:

$$F_m\mathcal{A}_{b,L} = \text{image}\left(\bigcup_{\ell\geq 0} x^{-\ell}\Psi_b^m(X;E)\right), \qquad m\in\mathbb{Z}.$$

The filtration is exhaustive, separated in the complete-symbol sense, and multiplicative. Principal symbols give

$$\operatorname{gr}_m \mathcal{A}_{b,L} \cong \mathcal{O}_L({}^bT^*X\setminus 0;\operatorname{End} E)_m,$$

the Laurent functions that are homogeneous of degree m in the fibers. In a b-Darboux chart, a choice of local Weyl quantization gives the expansion

$$a\#c \sim ac + \frac{1}{2i}\{a,c\}_b + \text{terms two orders lower}, \tag{4}$$

where changing quantization changes the individual lower-order symmetric terms. Without a chosen global symplectic connection, the full Weyl formula is not coordinate invariant. What is invariant, and all that is used below, is the commutator symbol

$$\sigma_{m+m'-1}([A,C]) = i^{-1}\{a,c\}_b.$$

This one-order drop is the origin of the first nonzero differential.

Lemma 4.1 (Morita reduction)**.** *The coefficient bundle E does not change the Hochschild homology.*

Proof. The global module of complete symbols with coefficients in E is a finitely generated projective equivalence bimodule between the scalar algebra and its End E-coefficient algebra. The associated matrix-trace and corner maps are continuous and filtration preserving, and the standard matrix-unit homotopy proves Morita invariance. This is a global Morita equivalence; no partition of unity is used to glue local chain homotopies. The topologically filtered version, including graded bundles, is treated in [2, 1]. □

We therefore suppress E. The Hochschild–Kostant–Rosenberg map on the associated graded algebra is

$$\chi(a_0\otimes\cdots\otimes a_j) = \frac{1}{j!}a_0\,da_1\wedge\cdots\wedge da_j.$$

For $w\in\mathbb{Z}$ let $\Omega_L^j({}^bT^*X\setminus 0)_{(w)}$ be the Laurent j-forms of fiber weight w, that is, with $\mathcal{L}_{\mathcal{E}}\alpha = w\alpha$ for the radial (Euler) vector field $\mathcal{E}$, and put $\Omega_L^j({}^bT^*X\setminus 0)_{\hom} = \bigoplus_w \Omega_L^j({}^bT^*X\setminus 0)_{(w)}$. If a_i is homogeneous of degree m_i, then $a_0\,da_1\wedge\cdots\wedge da_j$ has weight $m_0+\cdots+m_j$, since d commutes with dilation.

Proposition 4.2 (The E^1-page)**.** *The HKR map identifies the first page of the order spectral sequence with homogeneous Laurent differential forms on ${}^bT^*X \setminus 0$, the fiber weight being the total symbol order:*

$$E^1_{p,q} \cong \Omega^{p+q}_L({}^bT^*X \setminus 0)_{(p)}.$$

Proof. On each b-cotangent chart the associated graded product is ordinary multiplication of scalar homogeneous symbols. Antisymmetrization gives the inverse to χ on Hochschild homology:

$$a_0\, da_1 \wedge \cdots \wedge da_j \longmapsto \sum_{\tau \in S_j} \operatorname{sgn}(\tau) a_0 \otimes a_{\tau(1)} \otimes \cdots \otimes a_{\tau(j)}.$$

The displayed antisymmetrization is not a literal chain-level inverse; the two composites become inverse after the Koszul homotopy. For the topologically filtered Laurent groupoid algebra, continuity, completion, and globalization are the HKR proposition used by Benameur–Nistor [2, 1]. It may be proved through the sheaf of local Koszul resolutions and its fine resolution, but a bare partition of unity does not patch chain homotopies. Fiber dilation commutes with the published quasi-isomorphism, so it retains homogeneity. A chain of total order p in Hochschild degree $k = p + q$ is sent to a k-form of weight p, which is the displayed indexing. □

5 The first differential is Poisson homology

For classical b-symbols a and c, symbolic composition gives

$$\sigma_{m+m'-1}([A, C]) = \frac{1}{i}\{a, c\}_b.$$

Consequently the first differential after HKR is, up to the harmless factor i, the Poisson differential

$$\delta_b = [\iota_{\pi_b}, d] = \iota_{\pi_b} d - d\iota_{\pi_b}.$$

Explicitly,

$$\begin{aligned}\delta_b(a_0\, da_1 \wedge \cdots \wedge da_j) = &\sum_{r=1}^{j} (-1)^{r+1} \{a_0, a_r\}_b\, da_1 \wedge \cdots \widehat{da_r} \cdots \wedge da_j \\ &+ \sum_{1 \le r < s \le j} (-1)^{r+s} a_0\, d\{a_r, a_s\}_b \wedge da_1 \wedge \cdots \widehat{da_r} \cdots \widehat{da_s} \cdots \wedge da_j. \end{aligned} \tag{5}$$

To see this differential directly, insert the order-one-lower part of (4) into each multiplication in the Hochschild boundary. The symmetric terms cancel after HKR antisymmetrization, while the commutator terms combine to (5).

Let $N = 2n$ be the dimension of ${}^bT^*X$, put $\mu_b = \omega_b^n/n!$, and let $\pi_b^\sharp : \Omega^j \to \Lambda^j T$ be contraction of every covector with π_b. Symplectic duality is the map

$$*_b : \Omega^j_L({}^bT^*X \setminus 0) \longrightarrow \Omega^{N-j}_L({}^bT^*X \setminus 0), \qquad *_b\alpha = \iota_{\pi_b^\sharp \alpha}\mu_b,$$

and $*_b^2 = 1$ up to the standard degree sign. Applying the two sides to a monomial in the Darboux b-frame

$$\frac{dx}{x}, dy_1, \ldots, dy_{n-1}, d\xi, d\eta_1, \ldots, d\eta_{n-1}$$

gives, with the displayed convention for δ_b,

$$*_b \delta_b = (-1)^{j+1} d *_b \quad \text{on } \Omega_L^j. \tag{6}$$

The calculation remains valid at $x = 0$: the displayed frame is smooth as a b-frame, and both operators preserve finite Laurent pole order. Therefore a δ_b-cycle of degree j is carried to a closed Laurent form of degree $2n - j$, while boundaries are carried to exact forms.

The operator $*_b$ also shifts fiber weight. Since α_b and hence ω_b have weight 1, μ_b has weight n and π_b has weight -1. Contracting a j-form of weight w with π_b in each slot gives a j-vector of weight $w - j$, and contracting this into μ_b gives

$$*_b : \Omega_L^j({}^bT^*X \setminus 0)_{(w)} \xrightarrow{\cong} \Omega_L^{2n-j}({}^bT^*X \setminus 0)_{(w+n-j)}. \tag{7}$$

In particular δ_b, which lowers both j and w by one, preserves $w - j$, consistent with d preserving weight.

Corollary 5.1. *On the homogeneous complex of the E^1-page, Poisson homology is Laurent de Rham cohomology with complementary degree and shifted weight:*

$$H_j^{\pi_b, L}({}^bT^*X \setminus 0)_{(w)} \cong H_L^{2n-j}({}^bT^*X \setminus 0)_{(w+n-j)}, \qquad H_j^{\pi_b, L}({}^bT^*X \setminus 0)_{\hom} \cong H_L^{2n-j}({}^bT^*X \setminus 0)_{\hom}.$$

6 Homogeneity and the symbol-order circle

Choose a positive fiber norm r on ${}^bT^*X \setminus 0$. Then

$${}^bT^*X \setminus 0 \cong Y \times \mathbb{R}_+, \qquad (q, r) \longmapsto rq.$$

By Proposition 4.2 and (7), the column $E^1_{p,\bullet}$ in Hochschild degree k corresponds under $*_b$ to $(2n - k)$-forms of weight $p + n - k$. This bookkeeping matters: δ_b lowers fiber homogeneity by one, whereas $*_b$ changes both form degree and homogeneity.

The zero-weight reduction is elementary. The Euler vector field $\mathcal{E} = r\partial_r$ is dilation invariant, so d and $\iota_{\mathcal{E}}$ preserve weight. If a form α has nonzero weight w, Cartan's formula gives

$$(d\iota_{\mathcal{E}} + \iota_{\mathcal{E}} d)\alpha = \mathcal{L}_{\mathcal{E}}\alpha = w\alpha.$$

Thus $w^{-1}\iota_{\mathcal{E}}$ is a contracting homotopy of the weight-w subcomplex, and only weight 0 contributes to $H_L^*({}^bT^*X \setminus 0)_{\hom}$. By (7), in Hochschild degree k the only surviving column is $p = k - n$, that is, $q = n$ (Section 8).

At weight zero, restriction to $r = 1$ gives a unique splitting

$$\alpha(r, q) = \alpha_0(q) + \frac{dr}{r} \wedge \alpha_1(q).$$

Since $d(dr/r) = 0$, its differential is

$$d\alpha = d_Y\alpha_0 - \frac{dr}{r} \wedge d_Y\alpha_1.$$

Thus the weight-zero complex is the direct sum of the de Rham complex of Y and a copy shifted by one degree. The complete-symbol order is an integer grading with no preferred logarithm;

cohomologically this two-term complex is canonically written as the de Rham complex of an auxiliary circle:

$$\alpha_0 + \frac{dr}{r} \wedge \alpha_1 \longleftrightarrow \alpha_0 + d\theta_\sigma \wedge \alpha_1.$$

The formal order variable dr/r is represented cohomologically by the generator $d\theta_\sigma$ of a circle. Hence

$$H^j_L({}^bT^*X \setminus 0)_{\hom=0} \cong H^j_L(Y \times S^1_\sigma). \tag{8}$$

This is the source of the extra S^1 in the answer; it is not a geometric circle in the fibers of ${}^bT^*X$, and no choice of fiber norm affects its cohomology class.

7 Laurent cohomology and the boundary face

Let V be a compact manifold with one embedded boundary hypersurface and defining function x. The Laurent de Rham complex is

$$\Omega^j_L(V) = \bigcup_{\ell \geq 0} x^{-\ell}\Omega^j(V), \qquad d : \Omega^j_L(V) \to \Omega^{j+1}_L(V).$$

Proposition 7.1 (Face decomposition)**.** *There is a split short exact sequence*

$$0 \to H^j(V) \to H^j_L(V) \xrightarrow{\text{Res}} H^{j-1}(\partial V) \to 0,$$

hence an isomorphism $H^j_L(V) \cong H^j(V) \oplus H^{j-1}(\partial V)$. A splitting sends $[\beta]$ to the class of $(-1)^{j-1}\widetilde{\beta} \wedge d(\chi \log x)$, where β is a closed $(j-1)$-form on ∂V, $\widetilde{\beta}$ is a collar extension, and χ is a cutoff supported in the collar and equal to one near the boundary.

Proof. Work in a collar and write a Laurent form uniquely as

$$\alpha(x) + \frac{dx}{x} \wedge \beta(x),$$

where the tangential forms $\alpha(x), \beta(x)$ are Laurent families with a finite principal part and a smooth Taylor tail. Filter the collar complex by pole order and then by normal jet. On its associated graded, a term has a definite normal weight q. For $q \neq 0$, Cartan's identity gives the contracting homotopy

$$h_q = q^{-1}\iota_{x\partial_x}, \qquad dh_q + h_q d = 1.$$

Induction over the finite negative weights removes the principal part; the remaining positive Taylor tail is smooth and belongs to the ordinary de Rham complex. This jet-filtration argument, rather than freezing all coefficients at $x = 0$, is the Laurent Poincaré lemma. To globalize, multiply the collar homotopies by a cutoff equal to one near ∂V; the error terms contain a derivative of the cutoff, are supported away from ∂V, and are therefore smooth. The only new normal class is the weight-zero generator dx/x.

In particular, after writing

$$\alpha = \alpha_0 + \frac{dx}{x} \wedge \beta_0 + \text{terms of nonzero normal weight},$$

the residue is

$$\text{Res}_{\partial V} : \Omega^j_L(V) \longrightarrow \Omega^{j-1}(\partial V), \qquad \text{Res}_{\partial V}(\alpha) = \beta_0|_{\partial V}.$$

A direct calculation in the displayed order gives

$$\operatorname{Res}(d\alpha) = -d_{\partial V} \operatorname{Res}(\alpha).$$

Thus residue is a chain map to the degree-shifted boundary complex. The Laurent Poincaré lemma identifies its homotopy kernel with $\Omega^*(V)$, yielding the quasi-isomorphism

$$\Omega^*_L(V) \simeq \Omega^*(V) \oplus \Omega^{*-1}(\partial V).$$

If β is closed of degree $j-1$, let $p : [0,\varepsilon)_x \times \partial V \to \partial V$ be the collar projection. Choose χ with support in this collar and choose $\widetilde{\beta} = p^*\beta$ on an open set containing $\operatorname{supp} d(\chi \log x)$. Extend it arbitrarily beyond that set. Then $d\widetilde{\beta} = 0$ everywhere that $d(\chi \log x)$ is nonzero, so the representative below is closed. This collar construction gives the cohomological splitting

$$\beta \longmapsto (-1)^{j-1}\widetilde{\beta} \wedge d(\chi \log x).$$

Its residue is β, because $\widetilde{\beta} \wedge dx/x = (-1)^{j-1}(dx/x) \wedge \widetilde{\beta}$ near ∂V. If $x' = e^f x$, then $d \log x' = d \log x + df$; changing the defining function, collar, or cutoff changes this splitting by an ordinary/exact correction. Hence the direct sum is natural as a filtered extension but its explicit vector-space splitting is not canonical. □

Discussion. **Where do higher poles go?** A pole such as $x^{-q-1}dx$, $q > 0$, looks more singular than dx/x, but it is $-q^{-1}d(x^{-q})$. Higher Laurent coefficients occur in the complex and are essential for closure under composition; they add no new normal cohomology. Only the logarithmic coefficient survives.

Apply Proposition 7.1 to $V = Y \times S^1_\sigma$, whose boundary is $Z \times S^1_\sigma$. Then use

$$H^r(W \times S^1) \cong H^r(W) \oplus H^{r-1}(W).$$

For $r = 2n - k$ this gives exactly the four summands in (2).

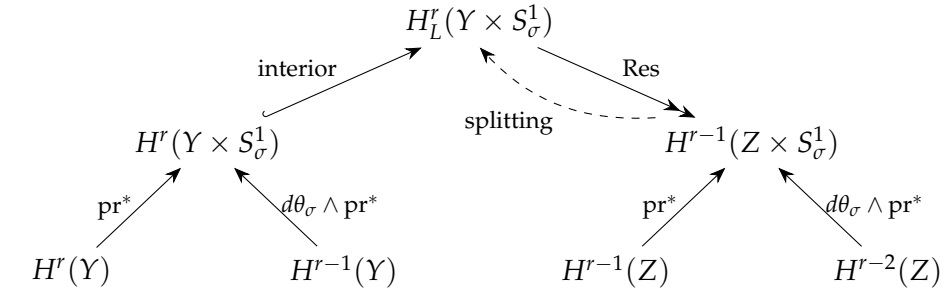


Figure 2: The four summands of (2), with $r = 2n - k$. The top row is the split exact sequence of Proposition 7.1 for $V = Y \times S^1_\sigma$; only the dashed splitting depends on choices. The bottom row is the Künneth decomposition for S^1_σ. The four summands come from the ordinary class, the symbol-order class $d\theta_\sigma \leftrightarrow d \log r$, the boundary logarithm $d \log x$, and their product.

For a manifold with corners, the same local argument has one generator $d \log x_H$ for every boundary hypersurface H through a point. If $\mathcal{F}_j(V)$ denotes the codimension-j faces, then

$$H^r_L(V) \cong \bigoplus_{j \geq 0} \bigoplus_{F \in \mathcal{F}_j(V)} H^{r-j}(F; o(N_F)). \tag{9}$$

Here $o(N_F)$ is the orientation line of the conormal directions to F. If the hypersurfaces have been globally cooriented and ordered, these lines are trivialized and the untwisted formula results. Indeed, apply the one-face residue successively. Residues for distinct hypersurfaces anticommute with the graded sign, so the summand attached to $F = H_1 \cap \cdots \cap H_j$ is represented by

$$\widetilde{\gamma} \wedge d\log x_{H_1} \wedge \cdots \wedge d\log x_{H_j}, \qquad [\gamma] \in H^{r-j}(F).$$

The product of the nonzero-weight contractions gives an acyclic complement. Changing the order of the hypersurfaces changes the orientation sign of the exterior logarithmic factors; the local system $o(N_F)$ records exactly this ambiguity. Iterated collar extensions provide a noncanonical splitting. This is only a sketch of (9): the Laurent Poincaré lemma above, applied one hypersurface at a time, reduces $\Omega^*_L(V)$ to the logarithmic (b-) de Rham complex, and (9) is then the standard b-de Rham theorem for manifolds with corners (for one boundary hypersurface see [7]; the corner case follows by iterating it over the boundary faces). The corner formula is not used elsewhere in these notes.

8 Collapse and assembly of the computation

Combining HKR, the Poisson differential, symplectic duality, and homogeneity gives the second page. The new point of this section is that the degeneration of the spectral sequence is then a matter of bidegrees; the only global input that remains is convergence.

Proposition 8.1 (The E^2-page and degeneration)**.** *For the order spectral sequence of $C^{\mathrm{cont}}_\bullet(\mathcal{A}_{b,L})$,*

$$E^2_{p,q} \cong \begin{cases} H_L^{2n-p-q}(Y \times S^1_\sigma), & q = n, \\ 0, & q \neq n. \end{cases}$$

Consequently $d^r = 0$ for all $r \geq 2$ and $E^2 = E^r$ for every $r \geq 2$.

Proof. By Proposition 4.2, $E^1_{p,q}$ is the space of Laurent k-forms of weight p, $k = p + q$, and $d^1 = i^{-1}\delta_b$ by Section 5. By (6) and (7), $*_b$ identifies $E^2_{p,q}$ with the cohomology of the Laurent de Rham complex in degree $2n - k$ and weight $p + n - k = n - q$. By Section 6, nonzero weights are acyclic, so $E^2_{p,q} = 0$ unless $q = n$; for $q = n$ the weight is zero and (8) gives $H_L^{2n-k}(Y \times S^1_\sigma)$. The differential d^r maps $E^r_{p,q}$ to $E^r_{p-r,q+r-1}$. For $r \geq 2$ the rows q and $q + r - 1$ are distinct, so at least one of source and target vanishes, and $d^r = 0$. □

The degeneration is therefore a consequence of degree counting once the weight bookkeeping (7) is in place. What is not automatic, and what we quote, is that the spectral sequence computes the continuous Hochschild homology at all.

Theorem 8.2 (Benameur–Nistor convergence theorem; quoted)**.** *For the topologically filtered Laurent complete-symbol algebra of a differential groupoid satisfying the hypotheses of [2, 1], the continuous HKR map identifies the E^1-page as in Proposition 4.2, and the order spectral sequence of the continuous Hochschild complex converges to $\mathrm{HH}^{\mathrm{cont}}_*$: the induced filtration $\mathcal{F}_p \mathrm{HH}_k$ (the image of the homology of $\mathcal{F}_p C_k$) is exhaustive and Hausdorff, and $E^\infty_{p,q} \cong \mathcal{F}_p \mathrm{HH}_{p+q} / \mathcal{F}_{p-1} \mathrm{HH}_{p+q}$.*

Benameur and Nistor also prove degeneration at E^2; in the present setting this is Proposition 8.1 and need not be quoted.

Proposition 8.3 (Source dictionary)**.** *The algebra $\mathcal{A}_{b,L}(X; E)$ is the one-face b-groupoid specialization of the algebras treated by Benameur–Nistor.*

Proof. The dictionary has six entries.

Groupoid. The differential groupoid is $\mathcal{G}_b$, with unit space X and Lie algebroid ${}^bTX$.

Laurent multipliers. The coefficient ring is precisely $\mathcal{O}(X) = C^\infty(X)[x_H^{-1}]$.

Algebra and topology. The algebra is $\mathcal{O}(X)(\Psi^\infty(\mathcal{G}_b; E)/\Psi^{-\infty}(\mathcal{G}_b; E))$, equipped with the topologically filtered completion of [1]; its continuous Hochschild complex is (3).

Anchor condition. The relevant property of the Lie algebroid is that its anchor becomes an isomorphism of Lie algebroids once the boundary defining functions are inverted. For ${}^bTX$ this is Proposition 3.1(i). We do not use, and do not claim, any chain-level identification of b-symbols with ordinary symbols through the anchor; see the end of the proof of Proposition 3.1.

Local models and coefficients. Locally near ∂X the groupoid $\mathcal{G}_b$ is the product of the pair groupoid of the boundary chart with the b-groupoid of $[0, \varepsilon)$. This product structure is the kind of local model their hypotheses require; we have not compared it with their precise formulation beyond this. The global Morita lemma removes the coefficient bundle E.

Spectral sequence. Their filtered HKR theorem identifies E^1 (Proposition 4.2) and their convergence theorem identifies E^∞ with the graded continuous Hochschild homology (Theorem 8.2). The E^2-page and the degeneration are computed independently here (Proposition 8.1). □

Proof of Theorem 1.1. By Proposition 8.1, $E^\infty_{p,q} = E^2_{p,q}$ vanishes unless $q = n$. Fix k. By Theorem 8.2, $\mathcal{F}_p \operatorname{HH}_k / \mathcal{F}_{p-1} \operatorname{HH}_k = 0$ for $p \neq k - n$. Hence $\mathcal{F}_p \operatorname{HH}_k = \mathcal{F}_{k-n} \operatorname{HH}_k$ for all $p \geq k - n$, and exhaustiveness gives $\mathcal{F}_{k-n} \operatorname{HH}_k = \operatorname{HH}_k$. Likewise $\mathcal{F}_p \operatorname{HH}_k = \mathcal{F}_{k-n-1} \operatorname{HH}_k$ for all $p \leq k - n - 1$, and the Hausdorff property gives $\mathcal{F}_{k-n-1} \operatorname{HH}_k = 0$. Therefore

$$\operatorname{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) = \mathcal{F}_{k-n} \operatorname{HH}_k \cong E^\infty_{k-n,n} = E^2_{k-n,n} \cong H_L^{2n-k}(Y \times S^1_\sigma),$$

with no choices involved: every map in this chain is canonical, and the identification (8) does not depend on the fiber norm. A Borel quantization, which splits the symbol filtration on the algebra only noncanonically, is not needed. Finally, Proposition 7.1 and Künneth for S^1_σ give the four summands in (2); here the choice of collar splitting enters. □

Discussion. **Why is convergence not automatic?** For a merely algebraic union of symbol orders, an infinite correction can escape the complex and an E^∞-class need not lift to a chain. Borel summation and the strict Laurent inductive limit are ingredients in the topologically filtered axioms: the former realizes asymptotic symbols and the latter keeps every individual chain at finite pole order. They do not by themselves prove convergence. Convergence to Hochschild homology is the quoted Theorem 8.2. The bidegree argument of Proposition 8.1 shows that no higher differentials occur, but by itself says nothing about whether E^∞ is the associated graded of $\operatorname{HH}_*^{\mathrm{cont}}$.

9 A Čech–de Rham route (conditional)

This section sketches an alternative organization of the proof, which replaces the global HKR assertion by a local calculation followed by descent. It is *not* self-contained: it depends on a quoted localization theorem for continuous Hochschild chains (Lemma 9.1), and it still uses the Benameur–Nistor convergence theorem; Čech descent alone computes the E^2-page, not the abutment.

Choose a finite good cover $\mathcal{U} = \{U_i\}$ of $Y = {}^bS^*X$. Near $Z = \partial Y$, take product sets $[0, \varepsilon)_x \times V_i$, and require every nonempty intersection to be contractible and collar adapted. Write $\widetilde{U}_I = U_I \times \mathbb{R}_+$ for the corresponding conic open set in ${}^bT^*X \setminus 0$.

Complete-symbol composition is local in ${}^bT^*X \setminus 0$, so for a conic open set $\widetilde{U}$ there is an algebra $\mathcal{A}_{b,L}(\widetilde{U})$ of Laurent complete symbols on $\widetilde{U}$, with restriction homomorphisms for $\widetilde{U}' \subset \widetilde{U}$. Let

$\mathcal{C}_q(U)$ be its Benameur–Nistor completed degree-q chain space: one forms the diagonal pieces $F^N_{N,N}$ over $\widetilde{U}$ first, takes their completed projective tensor powers, and only then takes the strict inductive limit in N, as in (3). Restriction of every tensor factor makes $U \mapsto \mathcal{C}_q(U)$ a presheaf of filtered complexes.

Form the double complex

$$K^{p,q} = \check{C}^p(\mathcal{U}, \mathcal{C}_q), \qquad D = \delta_{\check{C}} + (-1)^p b. \tag{10}$$

The total degree is $q - p$, since b lowers Hochschild degree and the Čech differential raises p.

Lemma 9.1 (Localization of continuous Hochschild chains; quoted input)**.** *The augmentation*

$$C^{\mathrm{cont}}_\bullet(\mathcal{A}_{b,L}) \longrightarrow \mathrm{Tot}\, K^{\bullet,\bullet}$$

is a filtration-preserving quasi-isomorphism, and it remains a quasi-isomorphism after passing to any fixed order quotient.

Remark 9.2 (Status of Lemma 9.1). The augmentation is *not* a quasi-isomorphism row by row: a chain $a_0 \otimes a_1$ whose factors are supported in disjoint open sets restricts to zero on every $\widetilde{U}_i$, although it is a nonzero chain. Such chains are, however, far from the diagonal, and the needed input is that continuous Hochschild chains localize to an arbitrarily small neighbourhood of the diagonal: the subcomplex of $\mathcal{U}$-small chains, all of whose factors are supported in a common $\widetilde{U}_i$, is a filtered deformation retract of the full complex, and on $\mathcal{U}$-small chains the Čech resolution is exact by a partition of unity. For the symbol algebra of a closed manifold, localization of this kind underlies the computation of Brylinski–Getzler [3]; for fine algebras it is Teleman's localization of the Hochschild complex [12]. We assume the corresponding statement for the topologically filtered Laurent algebra, compatibly with the order filtration; we do not prove it here.

Filter (10) by total symbol order. The associated graded of the total complex is the total complex of $\check{C}^p(\mathcal{U}, \mathrm{gr}\,\mathcal{C}_q)$, on which multiplication is commutative. Compute its homology by first taking Hochschild homology in each Čech degree. The local HKR map and the local Koszul homotopy, which commute with restriction, give

$$\check{C}^p\big(\mathcal{U}, \Omega^q_L({}^bT^*X \setminus 0)_{\mathrm{hom}}\big). \tag{11}$$

This is an intermediate page, not yet E^1 of the total complex: the Čech differential remains. Each weight-w piece of the sheaf of Laurent forms with a fixed pole bound is a module over $C^\infty(Y)$, hence fine, and the union over pole bounds is still Čech acyclic because multiplication by a partition of unity on Y preserves pole bounds and weight. Since the cover is finite, the Čech direction is bounded and (11) has Čech cohomology only in degree $p = 0$, equal to the global forms. Hence

$$E^1(\mathrm{Tot}\, K) \cong \Omega^*_L({}^bT^*X \setminus 0)_{\mathrm{hom}},$$

the same page as in Proposition 4.2. The first order-lowering part of the symbol product gives $d^1 = i^{-1}\delta_b$, exactly as in (5).

For a closed manifold these are exactly the local filtered steps in the Brylinski–Getzler calculation [3]: complete symbols pass to homogeneous functions, HKR gives forms, and the first quantum correction is the Poisson boundary. The present descent argument adds the Laurent normal-weight contraction, the residue complex at Z, and the strict topologically filtered completion required at the boundary.

Proposition 9.3 (Čech computation of the second page)**.** *Granting Lemma 9.1, the total filtered complex* (10) *has*

$$E^2_{p,q} \cong H_L^{2n-p-q}(Y \times S^1_\sigma; \mathbb{C}) \text{ for } q = n, \qquad E^2_{p,q} = 0 \text{ otherwise.}$$

This identification is independent of the good cover and the partition of unity.

Proof. Symplectic duality (6) converts the Poisson differential to the Laurent de Rham differential with complementary degree, with the weight shift (7). The Euler homotopy removes nonzero fiber weights. At weight zero a form is $\alpha_0 + (dr/r) \wedge \alpha_1$, so (8) identifies the remaining complex with Laurent forms on $Y \times S^1_\sigma$; the row $q = n$ is singled out as in Proposition 8.1. Refining the cover gives a common refinement, and the two refinement maps are homotopic on the augmented Čech resolution. This proves independence of all auxiliary choices. □

The boundary term can also be read directly in the descent complex. One must not write the generally false short exact sequence $0 \to \Omega^*(Y) \to \Omega^*_L(Y) \to \Omega^{*-1}(Z) \to 0$: a Laurent form with zero residue can still have higher poles. Instead let $\mathcal{K}^* = \ker(\mathrm{Res}_Z)$. The nonzero normal-weight contraction in the proof of Proposition 7.1 gives a quasi-isomorphism

$$\Omega^*(Y) \longrightarrow \mathcal{K}^*.$$

The exact sequence

$$0 \longrightarrow \mathcal{K}^* \longrightarrow \Omega^*_L(Y) \xrightarrow{\mathrm{Res}_Z} \big(\Omega^{*-1}(Z), -d_Z\big) \longrightarrow 0$$

therefore yields the derived decomposition

$$R\Gamma(Y; \Omega^*_L) \simeq R\Gamma(Y; \Omega^*) \oplus R\Gamma(Z; \Omega^{*-1}), \tag{12}$$

after choosing a collar splitting. This is precisely the interior/boundary decomposition used in (2); it also shows why the splitting, unlike the residue filtration, is not canonical.

Finally, Lemma 9.1 identifies the abutment of the order filtration on (10) with $\mathrm{HH}^{\mathrm{cont}}_*(\mathcal{A}_{b,L})$, compatibly with filtrations, so Theorem 8.2 applies to it; the proposition identifies its second page, and the bidegree argument of Proposition 8.1 gives $E^2 = E^\infty$. This recovers Theorem 1.1, conditionally on the localization input. Descent replaces the globalization part of the HKR step, but it does not manufacture convergence.

10 A Rees–Fedosov reorganization

This section makes the deformation behind symbol composition explicit. It is a heuristic reorganization of the computation, not an independent proof: nothing in it is used in the proof of Theorem 1.1, the E^2-page already follows from Sections 4–7, and the formal Hochschild homology computed below lacks the order circle. It has two logically separate parts: a formal Lie-algebroid formality statement and a finite-order Rees comparison with classical symbols, which is elementary.

This is also the deformation-theoretic refinement of the Brylinski–Getzler filtration: their principal-symbol and Poisson differentials are the first two coefficients of the Rees star product below [3].

10.1 The Rees deformation

The algebraic Rees algebra of the order filtration is

$$\mathcal{R}_F(\mathcal{A}_{b,L}) = \bigoplus_{m\in\mathbb{Z}} t^m F_m \mathcal{A}_{b,L} \subset \mathcal{A}_{b,L}[t, t^{-1}].$$

It records both specialization to the associated graded and the return to complete symbols at $t = 1$. Since $[F_m, F_{m'}] \subset F_{m+m'-1}$, its commutator is divisible by the Rees parameter in an associated-graded chart. Equivalently, for a symbol of fixed upper order $a \sim a_m + a_{m-1} + a_{m-2} + \cdots$, factor off the bookkeeping degree t^m and encode the descending expansion by

$$a_m + \hbar a_{m-1} + \hbar^2 a_{m-2} + \cdots . \tag{13}$$

The power t^m records the absolute order. After it is factored off in a fixed sector, we write the same Rees parameter as $\hbar$; its successive powers record loss of order. Thus (13) is an $\hbar$-adic chart on one upper-order sector, not a global $\mathbb{C}[[\hbar]]$-module structure on the union of all orders.

On Laurent functions on $M = {}^bT^*X \setminus 0$, full symbol composition in this chart becomes

$$a \star c = \sum_{j\geq 0} \hbar^j C_j(a, c), \qquad C_0(a, c) = ac, \tag{14}$$

where C_j is b-bidifferential and lowers the unscaled symbol order by j. Its antisymmetric first term is

$$C_1(a, c) - C_1(c, a) = i^{-1}\{a, c\}_b,$$

so $[a, c]_\star = (\hbar/i)\{a, c\}_b + O(\hbar^2)$.

The space M has the symplectic Lie algebroid ${}^bTM$ with symplectic form ω_b. Choose a torsion-free b-symplectic connection and use the homogeneous cotangent Fedosov construction, so that positive fiber dilation acts equivariantly. The Fedosov Weyl bundle has a flat differential of the form

$$D_F = -\delta + \nabla + \frac{i}{\hbar}\operatorname{ad}(r_F), \qquad D_F^2 = 0.$$

Flat sections, transported to Laurent functions by the Fedosov–Taylor map, should give a star product equivalent to (14). Fedosov quantization of symplectic Lie algebroids is developed by Nest–Tsygan [9]; ${}^bTM$ with ω_b is a symplectic Lie algebroid, but we have not checked the homogeneous version or the comparison with the b-symbol product in detail. Equivalent star products induce isomorphic Hochschild complexes, so the choice of connection does not affect the answer.

10.2 Hochschild-chain formality

Calaque–Dolgushev–Halbout prove formality theorems for Hochschild chains in the Lie-algebroid setting [4]; Dolgushev's smooth-manifold theorem is the corresponding prototype [5]. Their chain complexes are built from Lie-algebroid jets and polydifferential data; we have not shown that they agree with the Laurent chains used here, so the following is a heuristic. Twisting their module quasi-isomorphism by the Maurer–Cartan element defining (14) would give

$$(C_\bullet(\mathcal{O}_L(M)((\hbar)), \star), b_\star) \simeq (\Omega^\bullet_L(M)((\hbar)), \mathcal{L}_{\pi_\hbar}), \tag{15}$$

where $\pi_\hbar = \hbar\pi_b + O(\hbar^2)$ is the formal Poisson tensor and $\mathcal{L}_{\pi_\hbar} = [\iota_{\pi_\hbar}, d]$.

The formal tensor $\pi_\hbar$ is symplectic over $\mathbb{C}((\hbar))$; let $\omega_\hbar$ be its inverse. The formal symplectic star operator $*_{\omega_\hbar}$ satisfies the same Darboux calculation as (6) and converts $\mathcal{L}_{\pi_\hbar}$ to d, up to an invertible scalar and the degree sign. Therefore

$$\mathrm{HH}_k^{\mathbb{C}((\hbar))}\big(\mathcal{O}_L(M)((\hbar)),\star\big) \cong H_L^{2n-k}(M;\mathbb{C})((\hbar)). \tag{16}$$

The formality maps are b-polydifferential. Hence they preserve finite Laurent pole order: locally $x\partial_x(x^{-\ell}) = -\ell x^{-\ell}$, and tangential b-derivatives do not create a larger pole.

Equation (16) concerns the extended formal star algebra, before restriction to classical homogeneous expansions and before Borel realization. It must not be read as the complete-symbol answer. In particular, $H_L^*(M)$ itself has no order-circle factor: that factor appears only after the homogeneous order complex is formed.

Remark 10.1 (Where the order circle does, and does not, come from). The complex in (15) is relative to the base field $\mathbb{C}((\hbar))$. Thus $d\hbar = 0$, and it is incorrect to insert an additional class $d\hbar/\hbar$ into (16). The single class giving S^1_σ is the radial zero-weight class dr/r in Section 6.

One can instead compute absolute Hochschild homology over $\mathbb{C}$ for the Laurent extension of a Rees algebra. Then a class $d\log t$ can appear. The Rees grading ties the exponent of t to fiber homogeneity, so on the diagonal order complex $d\log t$ and $-d\log r$ describe the same order direction. After specialization at $t = 1$, only the radial class remains. Either convention yields one order circle, never two.

10.3 From formal series to classical complete symbols

Formal formality does not by itself compute the continuous homology of $\mathcal{A}_{b,L}$. What follows directly from the classical symbol calculus is a comparison at every finite depth of the order filtration.

Proposition 10.2 (Finite-order Rees comparison)**.** *After fixing a homogeneous classical symbol map, for every $m \in \mathbb{Z}$ and $N \geq 0$, the first $N+1$ symbol components give an isomorphism, continuous at every fixed pole bound,*

$$F_m\mathcal{A}_{b,L}/F_{m-N-1}\mathcal{A}_{b,L} \cong \bigoplus_{j=0}^{N} \mathcal{O}_L(M)_{m-j}. \tag{17}$$

Under (17)*, operator composition is the star product* (14) *modulo $\hbar^{N+1}$. These isomorphisms are compatible with truncation, completed projective tensor powers on each diagonal piece, and the Hochschild boundary. Consequently the Rees and classical order spectral sequences have the same E^1-page and the same Poisson differential d^1.*

Proof. There are three points to check.

Algebraic realization. Taylor expansion in symbol order gives (17); no infinite summation is needed at finite N. The b-symbol composition formula shows coefficient by coefficient that the two products agree modulo $\hbar^{N+1}$. Its first antisymmetric correction is the b-Poisson bracket, proving the assertion about d^1.

Topology. For a fixed Laurent bound, both sides of (17) are finite direct sums of nuclear Fréchet spaces. Truncation and the coefficient maps are continuous. Taking completed projective tensor powers first and the strict inductive limit over diagonal pieces afterward is exactly the order of operations in (3); hence the comparison extends to the finite quotients of the continuous Hochschild complex.

Inverse limit. Fix a pole bound ℓ. If $x^{-\ell}A \in F_{m-N-1}$ with $A \in \Psi_b^m$, then the full symbol of A has order $m-N-1$, so $x^{-\ell}\Psi_b^m \cap F_{m-N-1} = x^{-\ell}\Psi_b^{m-N-1}$. The b-symbol Borel lemma realizes every compatible sequence of truncations at this pole bound by a classical symbol, uniquely modulo smoothing; thus

$$\varprojlim_N x^{-\ell}\Psi_b^m / x^{-\ell}\Psi_b^{m-N-1} \cong x^{-\ell}\Psi_b^m / x^{-\ell}\Psi_b^{-\infty}.$$

The statement is false without the pole bound: an element of $\varprojlim_N F_m/F_{m-N-1}$ may need pole orders $\ell_N \to \infty$ as $N \to \infty$, and so need not come from F_m. This asymptotic completeness does not, by itself, prove that Hochschild homology commutes with the projective/inductive limits; that separate convergence assertion is exactly the topologically filtered theorem quoted in Section 8. □

Remark 10.3 (Scope of the Rees–Fedosov picture). Formality would organize the local differential beyond its first Poisson term, and Proposition 10.2 identifies the finite classical jets to which such a calculation applies. On the homogeneous Poisson complex the Euler contraction makes every nonzero weight acyclic, and the zero-weight splitting $\Omega_L^j(Y) \oplus (dr/r) \wedge \Omega_L^{j-1}(Y)$ has cohomology $H_L^j(Y \times S^1_\sigma)$. This reproduces the E^2-page of Proposition 8.1, but only because those steps are the ones already carried out in Sections 5–6; neither (15) nor (16) is used. The formal answer (16) is a different group (over $\mathbb{C}((\hbar))$, for all of M, without the order circle), and nothing here licenses specialization of formal Hochschild homology to continuous Hochschild homology. The Rees picture is therefore a heuristic explanation of where the Poisson differential comes from, not a second proof of Theorem 1.1.

$$\begin{array}{ccccc}
C_\bullet^{\mathrm{cont}}(\mathcal{A}_{b,L}) & \xrightarrow[\text{(localization)}]{\check{\text{C}}\text{ech}} & \mathrm{Tot}\,\check{\mathrm{C}}^\bullet(\mathcal{U}, \mathcal{C}_\bullet) & \xrightarrow[(E^2)]{\text{HKR, } \delta_b,\ *_b} & H_L^{2n-\bullet}(Y \times S^1_\sigma) \\
\Big\downarrow{\scriptstyle \text{Rees jets}} & & & & \\
C_\bullet(\mathcal{O}_L(M)((\hbar)), \star) & \xrightarrow[\text{formality}]{} & \left(\Omega_L^\bullet(M)((\hbar)), \mathcal{L}_{\pi_\hbar}\right) & \xrightarrow[*_{\omega_\hbar}]{} & H_L^{2n-\bullet}(M)((\hbar))
\end{array}$$

Figure 3: Top row: the Čech route globalizes the filtered HKR–Poisson calculation; it depends on the quoted localization Lemma 9.1 and on the quoted convergence theorem. Bottom row (dashed): the Rees–Fedosov picture, a heuristic only (Remark 10.3). Finite Rees jets compare classical symbols with the formal star product at each finite order, but the formal Hochschild homology lives over $\mathbb{C}((\hbar))$, on all of $M = {}^bT^*X \setminus 0$, and has no order circle; it is not mapped to the answer.

11 Checks and consequences

11.1 The closed-manifold case

If $\partial X = \varnothing$, Laurent cohomology is ordinary cohomology and $Z = \varnothing$. Formula (2) reduces to

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}(X;E)) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X),$$

equivalently $H^{2n-k}(S^*X \times S^1_\sigma)$. This is the Brylinski–Getzler computation [3].

11.2 Degree zero and residue traces

For the completed chain complex (3), degree-zero Hochschild homology is

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}) = \mathcal{A}/\operatorname{im}(b : C_1^{\mathrm{cont}} \to \mathcal{A}).$$

We abbreviate the image by $[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}$; it need not be closed. Its Hausdorffization is the different quotient

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A})_{\mathrm{H}} = \mathcal{A}/\overline{[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}}.$$

Theorem 1.1 computes the former group; by itself it does not prove that the commutator image is closed (but see below for $k=0$). Continuous traces factor through the Hausdorffization.

Now assume $n \geq 2$, X connected and $\partial X \neq \varnothing$. Then Y is connected, since its fibers S^{n-1} are connected, and Z has one component over each component of ∂X. The manifold Y is orientable, being diffeomorphic to S^*X (Proposition 3.1(ii)), a cooriented hypersurface in the symplectic manifold $T^*X \setminus 0$; hence its boundary Z is a closed orientable manifold of even dimension $2n-2$. (It is not a contact manifold, which would require odd dimension.) In (2) with $k=0$, $H^{2n}(Y)=0$ and $H^{2n-1}(Z)=0$ for dimension reasons, and $H^{2n-1}(Y)=0$ because Y is a connected compact manifold with nonempty boundary. Hence

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-2}(Z) \cong \mathbb{C}^{\pi_0(\partial X)}.$$

Continuous traces are continuous linear functionals vanishing on $[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}$, so they form a space of dimension at most $|\pi_0(\partial X)|$. Conversely, the boundary-face residue functionals, one for each component of ∂X (see [1]; compare the residue traces of the cusp calculus in [8]), are continuous traces; granting their linear independence, the space of continuous traces has dimension exactly $|\pi_0(\partial X)|$. In that case $[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}$ is the joint kernel of finitely many continuous functionals, so it is closed and $\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}) = \mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A})_{\mathrm{H}}$. If ∂X is connected, the trace is the b-residue trace up to scale.

11.3 A product cylinder

Let $X = [0,1] \times N$, where N is closed of dimension $n-1$, and put

$$W = S(\mathbb{R} \oplus T^*N).$$

Then $Y \simeq [0,1] \times W$ and $Z = W \sqcup W$. Therefore

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-k}(W) \oplus H^{2n-k-1}(W)^{\oplus 3} \oplus H^{2n-k-2}(W)^{\oplus 2}.$$

This example separates the order-circle contribution from the two boundary faces.

12 The unreduced operator algebra

Let

$$\Psi_{b,L}^{\mathbb{Z}}(X;E) = \bigcup_{m\in\mathbb{Z},\,\ell\geq 0} x^{-\ell}\Psi_b^m(X;E), \qquad \Psi_{b,L}^{-\infty}(X;E) = \bigcup_{\ell\geq 0} x^{-\ell}\Psi_b^{-\infty}(X;E),$$

the Laurent b-operators of integral order and their residual ideal (an algebra and a two-sided ideal, since conjugation by x^ℓ preserves each Ψ_b^m), each with a chosen topology. The unreduced operator algebra fits into

$$0 \longrightarrow \Psi_{b,L}^{-\infty}(X;E) \longrightarrow \Psi_{b,L}^{\mathbb{Z}}(X;E) \longrightarrow \mathcal{A}_{b,L}(X;E) \longrightarrow 0. \tag{18}$$

The answer for $\Psi^{\mathbb{Z}}_{b,L}$ is not obtained by simply reusing (1). It depends on the topology placed on the residual ideal and on whether kernels are required to vanish rapidly at the side faces, merely have polyhomogeneous expansions, or are completed in an operator norm.

For an H-unital residual ideal, excision gives the long exact sequence

$$\cdots \to \mathrm{HH}_k(\Psi^{-\infty}_{b,L}) \to \mathrm{HH}_k(\Psi^{\mathbb{Z}}_{b,L}) \to \mathrm{HH}_k(\mathcal{A}_{b,L}) \xrightarrow{\partial} \mathrm{HH}_{k-1}(\Psi^{-\infty}_{b,L}) \to \cdots .$$

Melrose and Nistor study the corresponding extension for the cusp calculus [8]. There the connecting map evaluates the index cocycle and produces an eta-invariant term. The analogous extension for the b-calculus is not computed in this paper; see also Moroianu–Nistor [11] for the relation between index and homology on manifolds with boundary.

Here H-unitality means that the completed bar complex of the nonunital residual ideal is acyclic; this is the hypothesis that prevents an additional excision defect. Continuous excision and H-unitality are substantial theorems of Wodzicki type [13]. Even for a rapidly decreasing nuclear smoothing algebra, H-unitality is not proved by a one-line finite-rank approximation; it has to be checked (or quoted) for the completion in use. Allowing polyhomogeneous side-face terms changes both topology and residual homology, so H-unitality must be verified anew for the chosen completion.

The Hochschild connecting morphism can be seen on a symbol cycle. Choose operator lifts of its tensor factors, apply the Hochschild boundary upstairs, and observe that the result lies in the residual ideal because its image vanishes in the quotient. The resulting residual homology class is independent of the lifts. This Hochschild map is not literally the Fredholm boundary map in K-theory. Compatibility appears only after applying the Chern character to cyclic homology and using excision in both theories. Boundary logarithmic classes then pair with regularized cyclic cocycles; the eta term is an analytic realization of such a pairing, not an element produced by the Hochschild boundary alone.

Thus (1) is the complete answer for Laurent complete symbols, while (18), together with the homology of the chosen residual ideal, is the correct computation for a specified unreduced operator algebra.

13 Relation to index theory

An elliptic b-operator determines an invertible matrix over the complete-symbol algebra. A Fredholm index additionally requires invertibility of its normal family. Its K_1-class has a Chern character in cyclic homology, which can pair with cyclic cocycles arising from the Hochschild classes above. In this qualified sense, the ordinary residue class supplies the interior symbolic term and a boundary logarithmic cocycle supplies the indicial correction. The operator-extension boundary map is the Fredholm index only under full ellipticity.

The work of Liu and Ma on differential K-theory and eta-invariant localization supplies a geometric refinement of this pairing: differential forms and eta data refine the topological K-class. It does not replace the Hochschild spectral-sequence calculation [6]. The logical division is therefore

Melrose–Nistor (cusp calculus) and Benameur–Nistor: algebraic homology,

Liu–Ma: differential-K and eta localization of index data.

Mazzeo–Melrose's fibred-boundary calculus gives complementary analytic background for normal families, mapping properties, and Fredholm criteria [10]. It does not replace the Benameur–Nistor topologically filtered Hochschild theorem. This distinction is useful when transporting the present computation to edge or fibred-cusp settings.

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